

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
Environmental Research Building (ERB) 1104 (lab), 3155 (office)
1910 University Drive
Boise, ID 83725

(303) 859-3106

hpmarshall@boisestate.edu

<https://www.boisestate.edu/earth-cryogars/>

Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington, Thesis: *Wet snow densification*

Selected Professional Experience

- 08/2018-8/2021 Project Scientist, NASA SnowEx Mission
- 08/2016-07/2018 Co-lead of Field Campaign, NASA SnowEx 2017 Mission
- 08/2022-present Professor, Department of Geosciences, Boise State University

Selected Relevant Peer-Reviewed Publications [h-index=33, i10-index=74, citations=3527]

Hoppinen, Z. M., Oveisgharan, S., **Marshall, H.P.**, Mower, R., Elder, K., and Vuyovich, C. (2024). Snow Water Equivalent Retrieval Over Idaho, Part B: Using L-band UAVSAR Repeat-Pass Interferometry. *The Cryosphere*, 18:575–592, doi.org/10.5194/tc-18-575-2024

Oveisgharan, S., Zinke, R., Hoppinen, Z., and **Marshall, H.P.** (2024). Snow Water Equivalent Retrieval Over Idaho, Part A: Using Sentinel-1 Repeat-Pass Interferometry. *The Cryosphere*, 18:559–574, doi.org/10.5194/tc-18-559-2024

Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with L-band InSAR. *The Cryosphere*, 17(5):1997–2019, doi.org/10.5194/tc-17-1997-2023

Marshall, H.P., Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

Rudisill, W., Flores, A., **Marshall, H.P.**, Siirila-Woodburn, E., Feldman, D., Rhoades, A., Xu, Z., and Morales, A. (2024). Cold-season precipitation sensitivity to microphysical parameterizations: Hydrologic evaluations leveraging snow lidar datasets. *Journal of Hydrometeorology*, 25(1):143–160

Hedrick, A., Marks, D., Kormos, P., **Marshall, H.P.**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063, doi.org/10.1029/2018WR023190

Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., **Marshall, H.P.**, Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996